

LOCAL PROTEIN DESIGN AND AGENT ORCHESTRATION

A Unified Framework for AutoGen, Ollama, AlphaFold, CollabFold, and RFdiffusion on Secure Mesh Networks

Author: Alexander Anderson

Compute Workstation: jb-gaming-pc (100.66.23.9)

Network Protocol: Tailscale Mesh VPN

Status: Production Deployment

Date: October 2023

1. Executive Summary

Alexander Anderson's local protein design and agent orchestration platform represents a major shift in computer-aided structural biology. By integrating multi-agent conversational frameworks with localized large language models and state-of-the-art structural prediction and design suites, the pipeline automates the complex, multi-step process of de novo protein generation. This report details the key architectural components, security configurations, and remote access systems that form the foundation of this state-of-the-art platform.

2. Agent Orchestration Layer (AutoGen & Ollama)

At the center of the orchestration layer is Microsoft's AutoGen framework. AutoGen enables the creation of highly specialized, collaborative AI agents that work together to solve complex computational challenges. These agents are designed with specific roles, such as sequence generation, structural evaluation, and workflow debugging, and communicate using natural language protocols.

To ensure absolute data privacy, eliminate external API dependencies, and enable offline processing, the agents are powered by local large language models hosted via Ollama. Ollama serves as a lightweight, efficient runner for high-performance open-source models like Llama-3 and Mistral. By running these models locally, the system guarantees that proprietary amino acid sequences and experimental designs never leave the secure local environment.

3. High-Resolution Structure Prediction (AlphaFold & CollabFold)

Predicting the three-dimensional structure of newly designed amino acid sequences is a critical step in verifying their viability and function. The platform integrates DeepMind's AlphaFold, a breakthrough deep learning system that predicts protein structures with atomic accuracy directly from their primary sequences.

To accelerate this process and enable high-throughput structural screening, the pipeline leverages CollabFold. CollabFold replaces AlphaFold's traditional, computationally expensive genetic search tools with MMseqs2 (Many-against-Many sequence searching). This optimization reduces the prediction time for standard protein domains from hours to minutes, allowing the orchestrating agents to evaluate hundreds of candidate sequences in near-real-time.

4. Generative Protein Design (RFdiffusion)

For de novo protein generation, the platform utilizes RFdiffusion (RosettaFold Diffusion). RFdiffusion is a powerful generative model based on the principles of denoising diffusion, adapted specifically for protein structures. It allows the design of entirely novel protein backbones customized for specific target binding, enzyme active sites, or structural scaffolding.

The orchestration agents construct input scripts for RFdiffusion, specifying target parameters, binding surfaces, and structural constraints. The resulting backbone structures are then passed directly to CollabFold for sequence design and structural validation. This closed-loop system allows the agents to iteratively refine designs, optimizing for binding affinity and folding stability without human intervention.

5. Secure Remote Infrastructure (Tailscale & jb-gaming-pc)

Running deep learning suites like AlphaFold and RFdiffusion requires massive GPU compute resources. To meet these hardware demands, Alexander Anderson has configured a high-end compute node, designated as jb-gaming-pc. This workstation houses powerful NVIDIA graphics cards optimized for training and inference workloads.

To provide secure, seamless, and low-latency remote access to jb-gaming-pc without exposing sensitive ports to the public internet, the platform relies on Tailscale. Tailscale creates a secure, encrypted mesh network (a virtual private tailnet) connecting all of Alexander's devices. Within this secure network, jb-gaming-pc is assigned the stable IP address 100.66.23.9. This configuration allows the orchestrating agents running on any authorized client device to trigger, monitor, and retrieve heavy compute jobs on the workstation securely from any location.

6. Platform Architecture and Components Summary

Component	Framework Role	Compute Host	Resource Profile
AutoGen	Multi-agent system coordinator	Local Client Node	Low CPU / RAM
Ollama	Local LLM inference runner	jb-gaming-pc (100.66.23.9)	Medium GPU / VRAM
AlphaFold	High-accuracy structure prediction	jb-gaming-pc (100.66.23.9)	High GPU / VRAM
CollabFold	Accelerated structure prediction	jb-gaming-pc (100.66.23.9)	High GPU / VRAM
RFdiffusion	De novo structural design	jb-gaming-pc (100.66.23.9)	High GPU / VRAM
Tailscale	Secure encrypted mesh meshwork	All System Nodes	Minimal Overhead